

Diagnosis

Clinical

The definition of stages is clinical, with chronology beginning with the onset of a chancre (ulcer or erosion). Stages can overlap.

Secondary syphilis develops in approximately one-third of untreated patients, and tertiary syphilis develops in about 10%.

Screening tests should also be offered to all attendees at dermatovenereology/genitourinary medicine (GUM)/STI clinics, referred to hereafter as 'sexual health clinics'.

Table 1: Treatment of Syphilis in Adults

Early Syphilis

(Primary, Secondary, and Early Latent — acquired <1 year previously)

- **First-line therapy option:**
 - Benzathine penicillin G (BPG) 2.4 million units intramuscularly (IM) as a single dose (either one injection of 2.4 million units or 1.2 million units in each buttock) on day 1
- **Second-line therapy option (if BPG is not available):**
 - Procaine penicillin 600,000 units IM daily for 10–14 days
- **Bleeding disorders:**
 - Ceftriaxone 1 g intravenously (IV) daily for 10 days
 - Doxycycline 200 mg daily (100 mg twice daily or single 200 mg dose) orally for 14 days

- **Penicillin allergy or parenteral treatment refused:**
 - Doxycycline 200 mg daily (100 mg twice daily or single 200 mg dose) orally for 14 days

Late Latent Syphilis

(acquired ≥ 1 year previously or of unknown duration), Cardiovascular, and Gummatous Syphilis

- **First-line therapy option:**
 - BPG 2.4 million units IM weekly on day 1, 8, and 15 (three total doses)
- **Second-line therapy option (if BPG is not available):**
 - Procaine penicillin 600,000 units IM daily for 17–21 days
- **Penicillin allergy or parenteral treatment refused:**
 - Some specialists recommend penicillin desensitization due to limited evidence for non-penicillin regimens
 - Doxycycline 200 mg daily (100 mg twice daily or single 200 mg dose) orally for 21–28 days

Neurosyphilis, Ocular, and Auricular Syphilis

- **First-line therapy option:**
 - Benzyl penicillin 18–24 million units IV daily (3–4 million units every 4 hours) for 10–14 days
- **Second-line therapy options (if IV hospitalization not possible):**
 - Ceftriaxone 1–2 g IV daily for 10–14 days
 - OR Procaine penicillin 1.2–2.4 million units IM daily **plus** probenecid 500 mg four times daily, both for 10–14 days
- **Penicillin allergy:**
 - Penicillin desensitization followed by first-line regimen

Pregnancy

- **First-line therapy for early syphilis (acquired <1 year previously):**
 - BPG 2.4 million units IM as a single dose (or 1.2 million units in each buttock)
- **Second-line therapy (if BPG unavailable):**
 - Procaine penicillin 600,000 units IM daily for 10–14 days
- **Penicillin allergy:**
 - Penicillin desensitization followed by first-line regimen

HIV-Infected Patients

- Treatment of syphilis should follow the same guidelines as for non-HIV-infected patients.